

Brian Tran

Top Secret Security Clearance

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Education

University of Virginia

Fall 2024 - Spring 2028

Bachelor of Science in Computer Science | Minor in Applied Mathematics

GPA: 3.64

Relevant Courses: Software Development, Computer Systems, Machine Learning, Statistics, Linear Algebra

Skills

Languages: Java, Python, TypeScript/Javascript, C/C++ SQL, HTML/CSS, Go

Frameworks: React, Next.js, Django, Django REST Framework, FastAPI

Tools: AWS (EC2, S3, Lambda, CloudFront), Docker, GitHub Actions, PostgreSQL, Prisma, Git, Selenium, Cursor

Experience

Full-stack Intern

May 2026 – August 2026

North Point Technology, LLC

Reston, VA

- Owned the end-to-end design, development, and deployment of a Next.js/Prisma metrics and data-tracking platform with dynamic form builders and visualization dashboards used by 80+ employees.
- Automated build and deployment with GitHub Actions CI/CD pipelines, cutting steps and catching failing builds.
- Containerized services with Docker, standardizing environments across dev and prod, and deployed to AWS EC2.
- Architected a React single-page application with a serverless Python FastAPI backend integrating enterprise APIs; deployed on AWS S3/Cloudfront for file management and AWS Lambda for backend.
- Secured enterprise applications with Azure Microsoft Entra ID (MSAL), JWT verification, and role access control.

Software Engineer

Sep 2024 - Jan 2026

theCourseForum

Charlottesville, VA

- Maintained a Python/Django course review platform that supports 26,000+ students for course selection.
- Introduced course filtering system and behind high-traffic search features to reduce reducing query latency by 30% and container CPU usage by 10% and improve user experience and usability.

Software Engineer Intern

May 2024 – August 2024

North Point Technology, LLC

Reston, VA

- Built and validated an automated Python testing and data-collection pipeline (BeautifulSoup, Selenium) on AWS EC2, verifying listing accuracy across 8 government contractor sites daily and surfacing 200+ new listings over two months.
- Improved pipeline reliability against bot detection by implementing proxy rotation and browser behavior simulation.
- Built an automated email notification service to deliver newly detected listings without manual review.

Projects

Dependency Risk Identifier Cursor Extension

- Built and published a TypeScript extension that scans npm and pip dependencies for vulnerabilities using OSV.dev, npm Registry, and endoflife.date, surfacing results via a sidebar and package.json diagnostics.
- Utilized Cursor Agent to generate upgrade recommendations with structured prompt, validating AI-suggested versions against semver compatibility rules before presenting them.

Graph-Representation of Naval Systems

- Built a graph-based application modeling onboard computing infrastructure for U.S. Navy stakeholders using React, Django REST Framework, and PostgreSQL as part of a 5-person Agile team.
- Developed RESTful APIs and backend validation to handle incomplete data; presented to judges and earned 3rd place at UVA's First Year Engineering Expo.

MapReduce Framework

- Implemented a distributed MapReduce framework in C++ and Go using processes, threads, and inter-process communication to efficiently distribute map and reduce tasks across multiple worker nodes.
- Designed task scheduling, synchronization, and fault-handling mechanisms to coordinate concurrent workers, manage intermediate key-value data, and reliably aggregate results across the distributed system.

Community

Extracurriculars: Vietnamese Student Association, Pickleball Club, Filipino Student Association, Music Nerds Club

Achievements: Managed year-long membership engagement for 400 members in Taiwanese Student Association. Raised \$8,000 from sponsorships for high school hackathon.